

Research plan – Planning product releases in global multi-team agile development projects

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Abstract—Release planning is a challenging part of market-driven product development and a success factor in agile software development projects. Release planning in an environment where multiple globally distributed teams are developing the same product requires careful coordination of responsibilities and technical dependencies. Existing release planning research has concentrated on mathematical model-based big upfront planning, which conflicts with agile methods. The goals of this research are to describe the release planning state-of-art in scientific literature and in real globally distributed agile software development projects, and to build a model for efficient release planning in globally distributed agile software development projects.

I. INTRODUCTION

Planning the next product release is a critical success factor in agile software development projects [1]. Due to the wicked nature of the problem [2], release planning is also one of the most challenging and important parts of market-driven product development [3], [4]. Globally distributed teams increase the difficulty of planning in agile software development organizations, as agile planning is based on face-to-face communication. Immediately after the development of a software product starts the schedule starts to deviate from the original release plan. There are many reasons for this *schedule variability*, but one of the most important is the difficulty of estimating development effort [5], especially when a deeply interconnected product is developed over several months and simultaneously by multiple teams. Typically the intended users of a software product cannot reliably convey their requirements before they get some hands-on experience with the product [3]. This *requirements variability* problem was already identified by [6] (“do it twice”) and later on affected the inception of rapid prototyping [7], [8] and agile software development methods [9], [10], [11]. In a stereotypical waterfall project the variability in schedule does not become apparent until late in the project and the variability in requirements is only revealed after the development project has ended. Iterative and incremental development models such as Rational Unified Process [12] mitigate the schedule variability by dividing the project into relatively short increments, but do not directly address the variability in requirements. However, requirements variability

is not always detrimental to project success. Sometimes the most uncertain requirements can provide the biggest value [13].

One proposed solution for unwanted requirements variability is to divide the development project into several incremental release projects. Each incremental release project should produce a set of features which are integrated, tested and of sufficient quality. Reinertsen [13] describes several mathematical and practical reasons that explain why reduction of batch size (i.e. incremental releases) is so effective against unwanted variability. According to Reinertsen, reducing batch size reduces variability in the development work flow, accelerates feedback and reduces risk. Reinertsen also states that large batches inherently decrease motivation and the feeling of urgency, may lead to even larger batches later on, and limit the progress of the batch to the worst (i.e. last) element of the batch. The optimal cadence of public releases varies by application area, and in a case where the optimal time in between public releases is long the organization should use more frequent internal test releases to reduce variability and risk, and to gather feedback from internal stakeholders. Figure 1 illustrates the amount of uncertainty over a development project with (a) one release or with (b) several incremental releases. t_0 is the beginning of the project, and t_R and $t_{R1...R3}$ are releases.

The purpose of this research is to fill the theoretical void that exists between the release planning and agile software development research in globally distributed organizations. As described above, these subjects have been recently identified as important topics by the SE research community.

The research question of this research is

“How should software release project planning be performed in agile software development organizations where many globally distributed teams are developing the same product at the same time?”

The main goals of this proposed research are to

- 1) **produce a description of the state of the art in release planning** and
 - 2) **produce a model of efficient release planning practices that produce high quality release plans**
- in globally distributed multi-team software product develop-

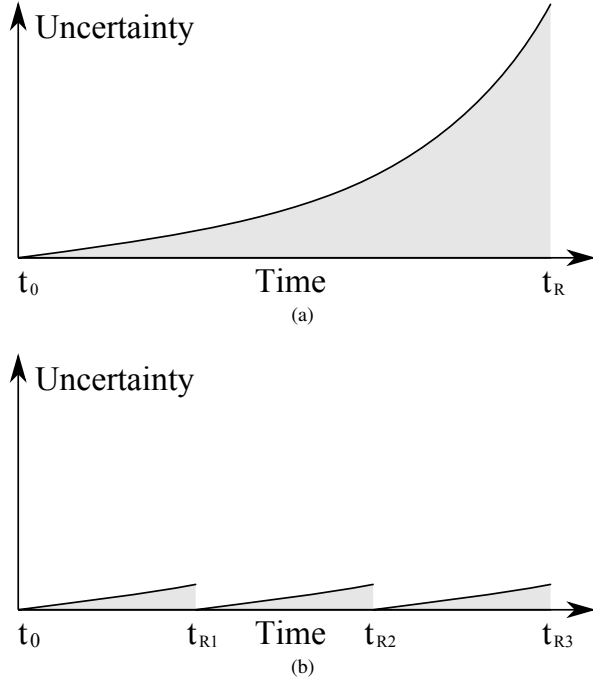


Figure 1. Uncertainty in (a) a single-release project and (b) a multiple-release project

ment projects which employ agile development methods.

II. RELATED WORK

The majority of publications on software release planning focus on different kinds of mathematical models and simulations designed to create the most valuable, satisfying or risk free release plans when key factors are known or can at least be estimated somewhat accurately [14]. The model or simulation is then used to generate one or a set of optimal or near-optimal release plan(s). Such models are called *strategic release planning models* [15]. To avoid confusion with strategic planning in NPD literature (see e.g. [16]), such models will be called mathematical release planning models (or mathematical models) in this research. Software evolvability has been linked to release planning: [17] describes how software anatomy can be used as an input for release planning, but does not elaborate on details of the release planning process.

The validity of most of the mathematical release planning models in a large-scale industrial setting is questionable. Svahnberg et al. [14] reviewed 22 mathematical release planning models. Only four of those models had been validated in large-scale industrial use. In addition, the model-driven approach to release planning has proven to be problematic in practice, as many software companies do not have a software development process which could be relied on to record or generate required key factors [18] and the factors might be context dependent and only discovered during release planning [2]. Many models are relatively complex and produce results that give impressions of false exactness and accuracy, which makes it difficult for practitioners to trust the results of such

models [2]. Often requirements change so frequently that any long-term plan quickly becomes obsolete [19], and the scalability problem¹ inherent to many requirements prioritization methods limits the scalability of the models. Agile software development methods claim to mitigate these issues by not creating detailed long-term plans and by adapting to changing requirements and priorities when needed [20].

Release planning in the agile software development context has only been slightly addressed in scientific publications. Our previous paper described how one organization performed release planning in distributed multi-team agile software development context [21]. The rest of the existing scientific publications [22], [23], [24] propose using a mathematical release planning model in an agile software development context. However, none of the mathematical models were empirically validated in real industrial use.

III. RESEARCH METHOD

The research will include both qualitative and quantitative methods. The primary research methods for the research are case study research [25] and action research [26]. According to [25], case study has distinct advantage over other research methods when “a ‘how’ or ‘why’ question is being asked about contemporary set of events over which the investigator has little or no control.” Release planning performed in software companies is very much contemporary and a researcher has practically no control over it. While researchers will not actively supervise or guide release planning, they will provide feedback and improvements suggestions to the companies. Thus the research must be also regarded as action research and analysis of the research must take into account all that entails [27]. Following a classification by [28], this action research can be described as “participant action research”. Data collection will be performed using several methods, which include observations, surveys, artifact analysis and interviews. The different data collection methods combined with the longitudinal and cyclical nature of release planning lends itself to triangulation, following the recommendations of [27]. According to [29] action research interventions in an organization must be based on solid theory which is improved on each cycle of the diagnose-plan-act-evaluate-reflect -process. Case study based theory building literature (e.g. [25], [30]) will provide additional guidance for theory building for building a solid qualitative theory for the research.

IV. RESULTS

This research has resulted in two scientific publications so far. The publications and their contribution to the research goals are described below.

- 1) “Rigorous Support for Flexible Planning of Product Releases - A Stakeholder-Centric Approach and its Initial Evaluation”, published in Proceedings of the Hawaii International Conference on System Sciences (HICSS), Jan 2010 [22].

¹For example, the number of comparisons in simple pairwise comparison is $O(n^2)$.

- 2) "A Revelatory Case Study on Scaling Agile Release Planning", published in Proceedings of the 36th Euromicro Conference on Software Engineering and Advanced Applications (Euromicro SEAA), 2010 [21].

Publication 1 presents a formalization of the release planning problem, describes a formal release planning process and gives a simple example of release planning in an agile software development context (goals 1 and 2). While the example use of the described release planning process is successful, the results show that the selection of planning factors and constraints must be conducted with great consideration to get the maximum benefit out of the process. Publication 2 contains a qualitative description of how a Finnish software product company performs agile release planning in a multi-team development project (goal 1). The main finding is that release planning can be performed in an agile way in a development organization of over ten development teams and total of 140 stakeholders.

V. EXPECTED RESULTS AND EVALUATION

Three more publications and a PhD thesis have been planned. The publications are described below. The conference articles will be (and have been) evaluated by the international research community in several high quality conferences, the journal articles will be evaluated by editors and reviewers, and the doctoral dissertation will be reviewed by experts in the preliminary examination and by the opponent in the public defense.

- 3) "Scaling Agile Release Planning - A Method and Descriptive Case Study", 34th International Conference on Software Engineering, The Software Engineering in Practice track (ICSE 2012), 2012.
- 4) "Managing Requirements in Globally Distributed Multi-Project Environment", Requirements Engineering journal, 2012.
- 5) "Planning Software Releases in Large Distributed Agile Development Projects", Empirical Software Engineering journal, 2013.
- 6) Doctoral dissertation, spring 2014.

Publication 3 describes an agile release planning method and a descriptive case study of its use. Publication 4 describes a case study on how requirements are managed in an agile organization that develops telecommunications software on multiple sites. Publication 5 describes a model for efficient release planning that produces high quality release plans in large-scale and multi-site agile software development organizations. Publication 6 is a doctoral dissertation which contains and summarizes all previous publications.

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